

Conversational Product Recommendation using LLM

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Abstract—It is expected to deploy chatbots as sales assistants on e-commerce platforms soon. In recent years, the capabilities demonstrated by large language models (LLMs) indicate their suitability for this role. However, a dialogue process is not fully established. Most existing works concentrate on searching for ideal products, such as Retrieval-Augmented Generation (RAG) and product search. Many proposed methods rely on product reviews. However, in the case of some e-commerce platform products, valuable reviews are often lacking, rendering these approaches impractical. Thus, we studied how to assist users and select a set of candidate products. Accordingly, we simulated sales conversations based on the product features provided by the seller, using LLM and reliable content while addressing the cold-start problem. Specifically, we used product features to generate customer persona for user simulation. LLM was used to play the role of sales assistant in giving recommendations. The experimental results showed that practical conversation was created between customers and chatbots in Traditional Chinese. The code and results are available at: https://github.com/terryobe-ncu/CPR_LLM

Keywords—conversational AI, eCommerce, dialog systems, large language model (LLM)

I. INTRODUCTION

Given the impressive conversational capability demonstrated by today's large language models (LLMs), we can deploy professional e-commerce sales assistant chatbots. Compared with conventional recommendation systems that unilaterally recommend products to customers, conversational recommender systems (CRSs) enable communication between systems and customers, allowing for more information exchange between both parties.

As e-commerce platforms develop, sales platforms categorize products in more detail, and the database contains more structured information. Retrieval augmented generation (RAG) is expected to serve as a more powerful product search engine [1]. In conversation, retrieval-based generation concepts are the key for LLM to avoid hallucinations, provide credible information, and improve scalability. Existing research on dialogues mainly involves collecting preferences for product searches in e-commerce [2–4]. Menu operations can essentially replace it. Therefore, Greater emphasis is placed on user needs and the persuasiveness of recommendations.

In this study, we used LLM as a user and sales assistant to generate conversations which is different from Javadi et al. [5], who proposed a review-based conversation generation approach. Taiwan e-commerce products often lack valuable reviews; most of the reviews describe "The quality is very good" or "Shipping is fast". Thus, the experiments in this study only used product features listed by sellers. Inspired by Wang et al., we used LLM to generate personas based on product features for user simulation [6]. The sales assistant recommended a certain product feature among the candidate products. The conversation continued until the sales assistant

mentioned the selected product feature or the round reached the limit.

We used Gemini-1.0-pro as the LLM model [7]. Gemini generates fairly fluent and realistic personas and the content has a certain degree of rewriting in wording between the references. Under the guidance, 70% of the conversations were delivered by the sales assistant. When users provide sufficient context and requirements, the sales assistant can provide recommendations that resonate with customers. We designed an LLM sales assistant system framework in which developers integrate product information or background knowledge to answer questions or provide professional advice.

II. DATA PREPROCESSING

We obtained the product information by crawling Taiwan's well-known MOMO e-commerce website using their defined categories [8]. We focused on 7 cosmetics and 8 health supplement keywords, as their effectiveness is more important than their mere appearances. For each of these fifteen keywords, we crawled data for products with the same functionality or target demographic. We grouped five products of different brands into a suggestion group. In MOMO's product information and descriptions, we ensured each good had exactly three descriptions as product features. At last, we had ten groups of data for each keyword. In total, there were 150 groups for 750 products. Table I shows an example of data for a product, along with the generated results.

TABLE I. SAMPLE DATA AND RESULTS

Field Names	Values
Product Types	<i>Skin toner</i>
Product Features	1. Reinforcement of healthy skin barrier 2. Self-stimulating moisturizing power 3. Concentrated moisturizing dry skin
Persona	"A 25-year-old office worker, Mei, whose skin is dry and rough due to long hours of late-night work, longs for a toner that strengthens the skin barrier and boosts moisturizing power so that her skin can regain a healthy glow. ° "
Codename with Target Feature ^a	A-1
Conversation	User: "I've been staying up late for a long time and my skin is very dry and rough. I'm looking for a toner that can strengthen the skin barrier and moisturize my skin, which one do you recommend?" Assistant: "We recommend [Product A], which features [A-1] to strengthen and protect the healthy skin barrier, and [A-2] to generate its moisturizing power, which is very suitable for your dry and rough skin. "

^aThis data is the first in the group, so its codename is "A"; the target is the first of product features.

III. FRAMEWORK AND EXPERIMENT

We divided the e-commerce system process into two stages to (1) collate a group of candidate products and (2) answer user questions or make recommendations. The model

recommended suitable items while extracting compelling product descriptions to captivate customers. The overview of this framework using LLM is shown in Fig. 1. In this regard, we experimented with crawled data to demonstrate the conversation scenarios. We chose the *Gemini-1.0-pro* as our

LLM model with excellent performance in Traditional Chinese and set the *temperature* to 0 to ensure deterministic results. Certain product features of Gemini resulted in the responses being blocked. To ensure the smooth operation, we disabled the relevant filters.

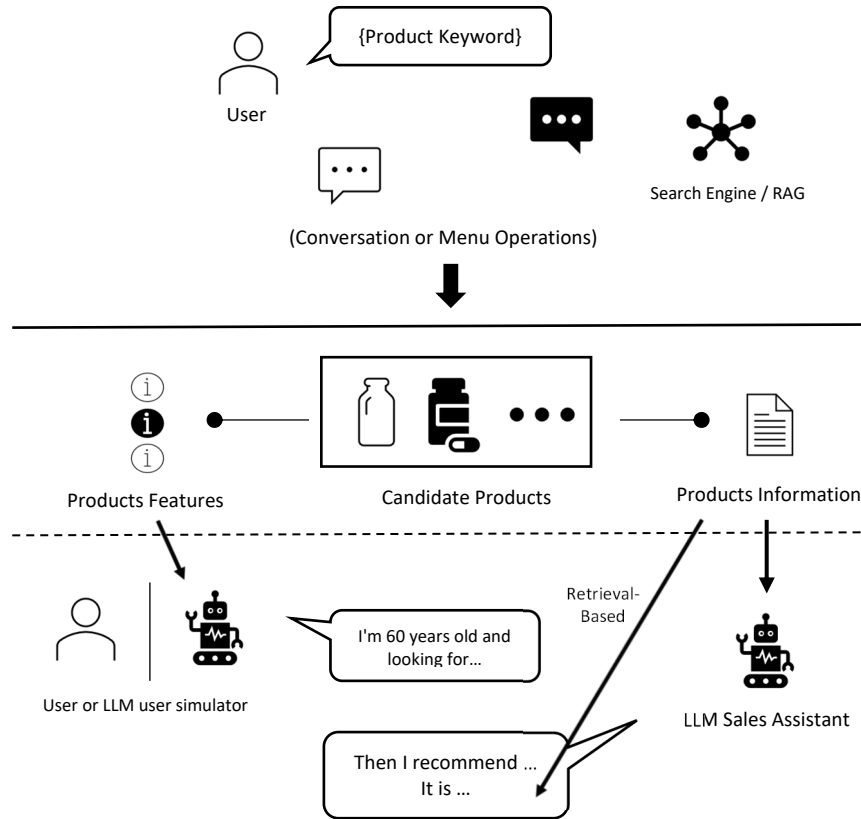


Fig. 1. Overview of LLM conversational product recommendation framework

A. User Simulation

To simulate, we created personas. Initially, we requested LLM to generate a fitting potential buyer's persona for each product based on its features. In the user simulation, LLM acted as that persona and interacted with the assistant to receive recommendations.

B. Target Selection

Different from most recommendation tasks that demand recommending a target item, the model recommended a product feature instead of a target item. To ensure the persuasiveness of our recommendations, the model mentioned the target as an objective indicator. For each group, we selected a product feature as the ground truth. Since the product features of the data contained promotional content, we used LLM to select one product feature for each group as a target. We expected the target to meet the following conditions so that the model prompted the LLM with such requirements:

- Clear and understandable, free from promotional content
- The feature should be most relevant to the customer's situation but differ in words.
- No other product offers a similar feature.

For each group, we listed all goods and their respective personas and requested LLM to select the feature from

corresponding codenames ("A-1" to "E-3"). We also asked LLM to provide further explanations for the reasons, guiding the model in considering them before responding.

C. Conversation Generation

The conversation started with a question from the user. We prepared an example in the API *message* to help control the style of user questions. However, since Gemini had no "system" role, this task was refined in in-context learning [7]. Specifically, we added a fixed prompt and response in the message's "model" and "user" roles, respectively, followed by the query. Once the user's first question was generated, the fixed prompt was replaced with another instruction. We instructed the simulated user to accept the salesperson's recommendation of the target product and inquire about other features of that product. If the salesperson recommended other products, the simulated user refused until the target was recommended.

For the sales assistant, we instructed LLM to suggest a product and highlighted the product features most relevant to the customer's situation. We used a code name to provide this information to the sales assistant. In the same way, we programmed a fake response for the AI model to agree with the instruction before we sent the user's first question as a prompt. These relevant product features were added in the initial sales assistant instruction but were not disclosed to the user. The conversation was terminated if the sales assistant mentioned the target feature. During development and testing,

both parties read the dialogue. The codename was converted into actual content to reveal the product features to the user.

IV. RESULTS

Through prompts, Gemini generated excellent personas. In unspecified circumstances, the model produced fluent personas in Traditional Chinese that aligned with Taiwanese linguistic conventions with no instances of Simplified Chinese observed.

For target selection, the platform name "MOMO" appeared 13 times in the product features but did not appear in any target features. However, we encountered the position bias. Table II shows the statistics of goods selection. Most selections were "A," followed by "C" so the distribution was not uniform. We tested the model by reversing the input content, but the last option was not frequent. This confirmed the presence of position bias in the model.

TABLE II. STATISTICS OF TARGET GOODS SELECTION

Target Goods Codenames	Original	Reversed
A	<u>116 (77.33%)</u>	<u>102 (68.00%)</u>
B	4 (2.66%)	6 (4.00%)
C	<u>26 (17.33%)</u>	<u>39 (26.00%)</u>
D	3 (2.00%)	3 (2.00%)
E	1 (0.66%)	0 (0.00%)

Most features were appropriated as targets except for those containing promotional content. The system prioritized selecting options that ranked higher in the list, but it did not explain the overlooked "B" option. Table III shows that there is also a slight position bias in the selection of target features, but it is important to note that the contents of product features are related to the platform. For example, if sellers tended to place promotional content in a specific position, the proportion of that position becoming a target decreased. Even so, the target selection was not critical in the experiment. We did not shuffle the order of items. Typically, there was a recommendation ranking when producing candidate products, and system developers can leverage the characteristics of position bias to make items that are originally ranked higher more likely to be recommended.

TABLE III. STATISTICS OF TARGET FEATURES SELECTION

Target Features	Original	Reversed
Feature 1	56 (37.33%)	63 (42.00%)
Feature 2	58 (38.66%)	54 (36.00%)
Feature 3	36 (24.00%)	33 (22.00%)

In the developed conversation model, the first user's question was generated using the persona. The questions generated by the model were considered actual examples. The instructions and the prepended example effectively provided concise descriptions based on the persona. The system used more colloquial wording rather than simply repeating the persona's text. For the sales assistant, the model followed instructions for using product codenames in the responses. However, it occasionally listed multiple product features at once or still included most of the text from those features even after using the codename. System adjustments and careful instructions are demanded to improve the quality of the

subsequent conversation. We analyzed the number of dialogue rounds generated in the current experiment. Table IV shows that nearly 70% of conversations ended in the first round, meaning the sales assistant mentioned the target feature's codename in the first reply.

TABLE IV. STATISTICS OF CONVERSATION ROUNDS

Rounds	1	2	3	4	5	Fail ^a
Number of Cases	103 (68.67%)	31 (20.67%)	7 (4.67%)	2 (1.33%)	1 (0.67%)	6 (4.00%)

^aThe sales assistant didn't mention the target feature after 5 rounds.

In selecting a feature randomly, the probability of correctness in the first round was 6.67%. The sales assistant output two features in 20% of the cases in the first round. The remaining 80% followed the instructions of one feature. This indicates that the LLM sales assistant effectively recommended suitable products if the customers described their situation. For the conversations with 2 rounds, about 67.74% of cases proposed the target goods in the first reply, but the mentioned features were not the target. Table V shows the average number of target feature types proposed in all replies from the sales assistant except the last one, as well as the total number and proportion that mentioned the target goods.

TABLE V. STATISTIC OF ASSISTANT'S REPLIES

Rounds	Average Number of Mentioned Features	Number of Case Mentioned the Target Good
1 ^a	-	-
2	1.06	21 (67.74%)
3	2.00	4 (57.14%)
4	3.00	2 (100.00%)
5	6.00	1 (100.00%)
Fail	2.50	2 (33.33%)

^aIf a conversation has only one round, excluding the last round renders the replies empty.

When the user's question was different from the persona in wording, it led the sales assistant to recommend items that were more suitable targets than the ground truth. In a few failed cases, users mistakenly accepted items that were not the target, leading the sales assistant to continue introducing the wrong item. This needs improvement in the multi-turn dialogue.

V. CONCLUSION

A novel LLM was developed as a sales assistant and validated for its feasibility. LLM generated personas of e-commerce customers and simulated user behavior for the first time. This approach solves the cold-start problem and can be easily implemented by any company or researcher. The retrieval-based replies are crucial for avoiding hallucinations and enhancing persuasiveness. However, further experiments are necessary to improve the professionalism of conversation by incorporating real-life experiences and domain knowledge. Safety settings and content moderation of LLMs are also important considerations for businesses to prevent presenting harmful content to customers. Clarifying question generation is also essential. e-commerce platforms can assist vendors or individual sellers in creating customer service chatbots.

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